

DESIGN

Lecture 23.

Lifting and control Surfaces structural design II: Torsion Box

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Introduction

Torsion Box Design

Skins

Sub-Structure

Spars

Ribs

Interfaces

Horizontal & Vertical Stabilizers

Introduction

Empennage structure has evolve as the wing has done.

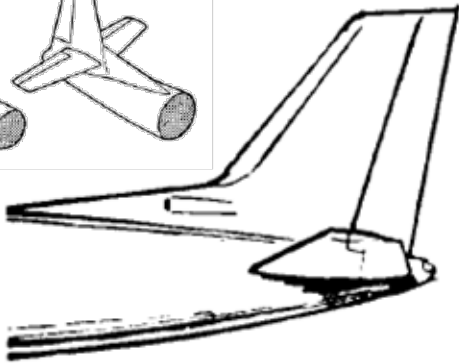
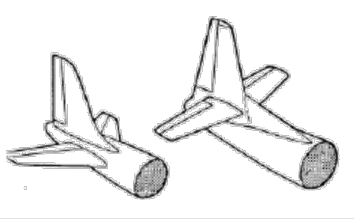
The aspect ratio of a vertical or horizontal surface usually tends to be smaller than a wing one. It implies lower bending moment.

The type of construction we can find in the stabilizer and fin are usually similar to the type of wing constructions:

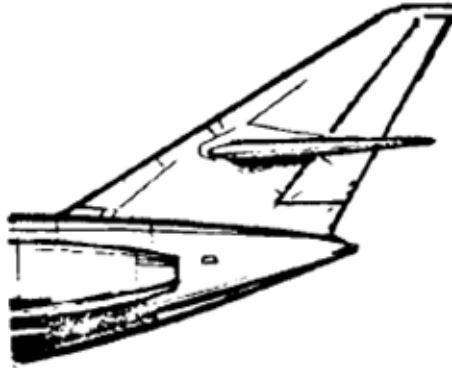
- **Single Spar** construction with **auxiliary rear spar** and all bending material concentrated in the **spar cap**. This configuration is typical of light aircraft construction.
- **Single Spar** with **Pivot at Root**. It allows to rotate the complete surface. This construction has been applied on modern high speed fighters (F18, Eurofighters,...)
- **Two-spar** construction with all bending material concentrated in **spar caps** or in the **surface panels**.
- **Multi-Spar** construction with spars supporting all the bending loads

Introduction

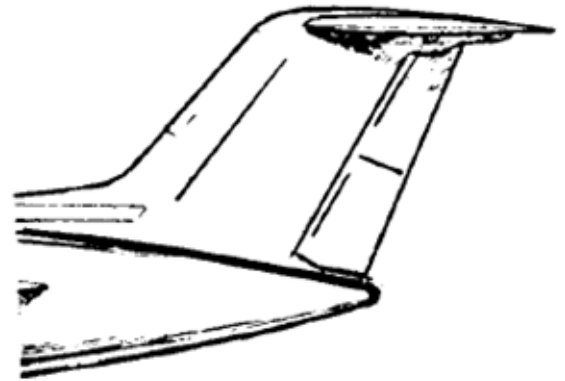
Following figures show the typical arrangement of transport tail.



Conventional Tail



Cross Tail

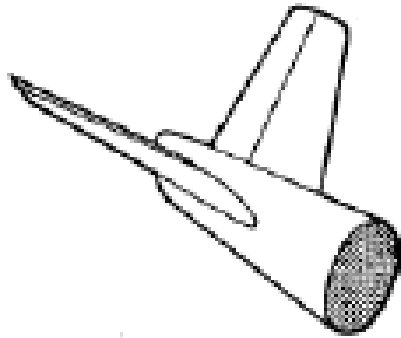


T Tail

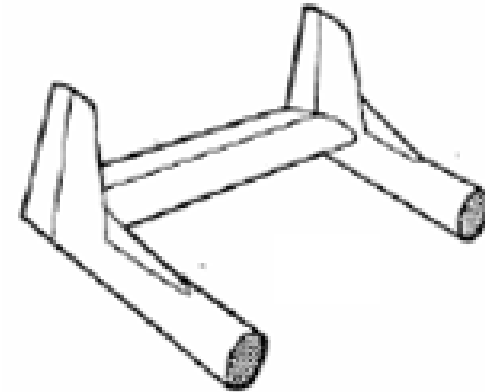


Introduction

Other Configurations.-



V Tail or Butterfly Tail



Twin Vertical Tail



Introduction

Other Configurations.-



Double Fin
(Typical Configuration for carrier
capable Aircrafts)

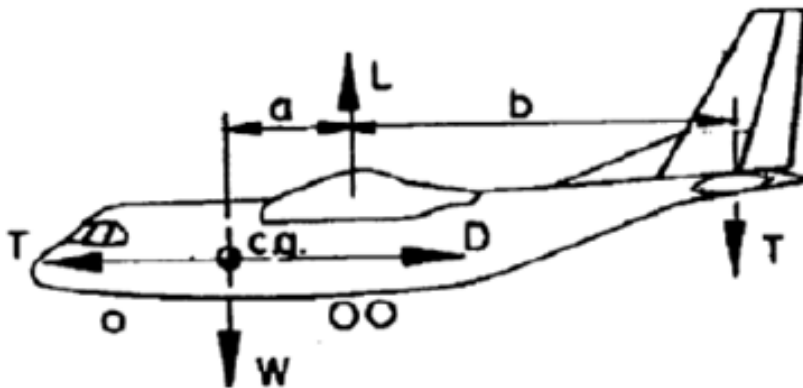


Single Fin & Canard

Loads

Empennage loads are described as follow:

- **Horizontal Tail** is put on airplane for two primary reasons:
 - Balance the moments caused by aerodynamic and inertia forces of other aircraft parts.
 - Provide control about pitching axis.
- Loads on the **Vertical Tail** are caused by:
 - Rudder deflection
 - Aileron deflection
 - Lateral gust
 - Asymmetric engine thrust

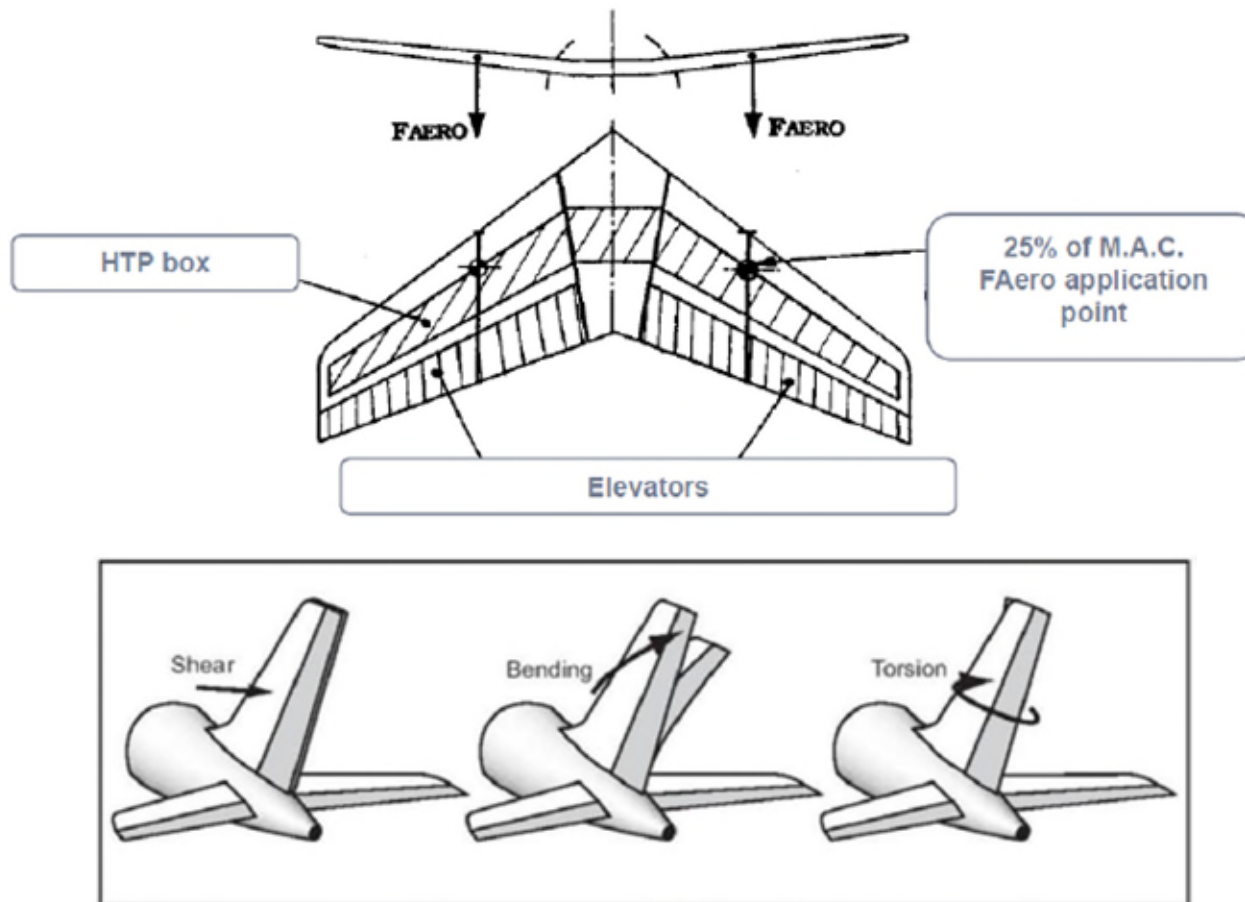


$$\sum F = 0$$

$$\sum M = 0$$

$$\left. \begin{aligned} T &= D \\ W &= L - T \\ T(a + b) &= L a \end{aligned} \right\}$$

Loads



Horizontal Tail

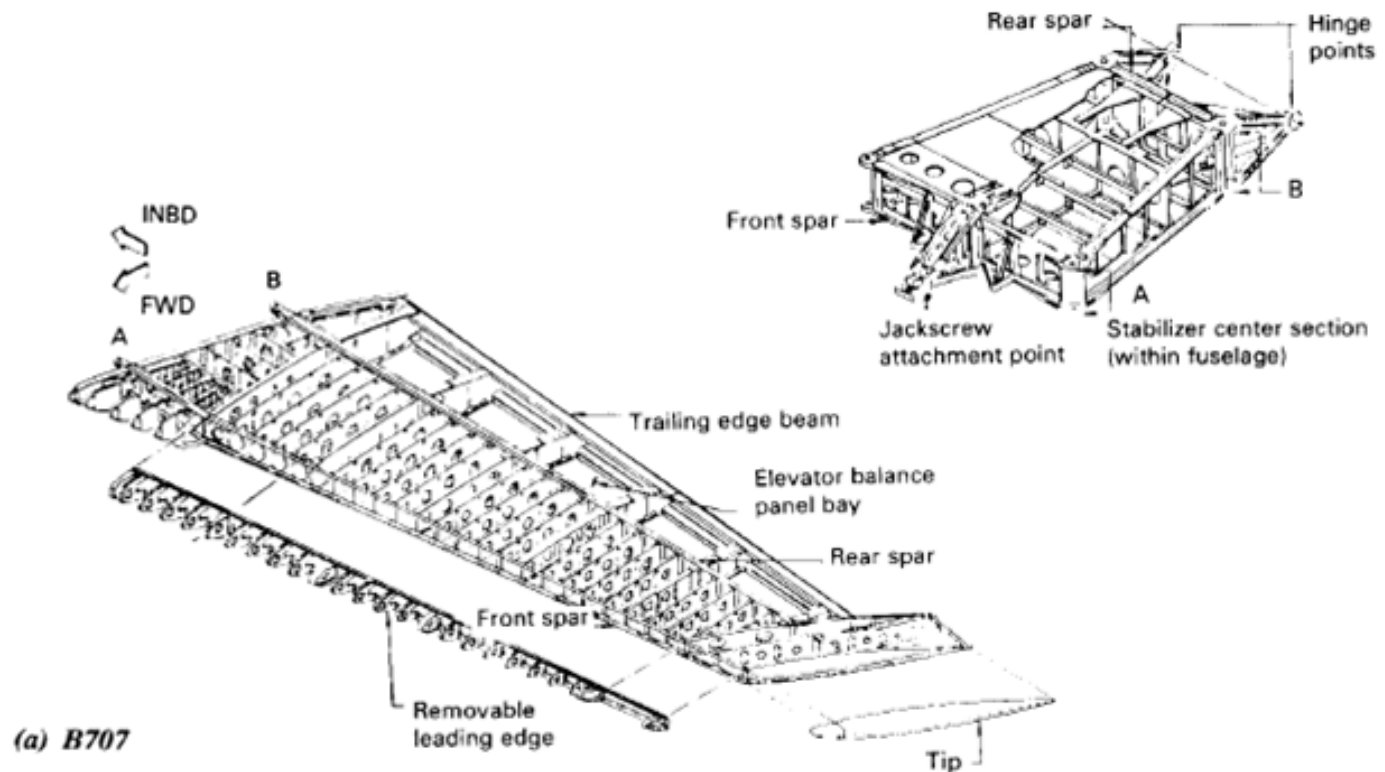
Typical Features

- Horizontal Stabilizer is usually a two-spar structure consisting of a center structural box section and two outer sections.
- The stabilizer assembly is interchangeable as a unit at the fuselage attach points
- The outer sections are interchangeable at the attachment to the center box
- Access Doors must be considered in the design of the perimetral structural elements (Front Spar Web, Rear Spar Web, ...) for inspection and maintenance of internal structure.
- Usually stabilizer assemblies are weather sealed with drain holes on the lower surface to prevent moisture retention.

Transport Configurations

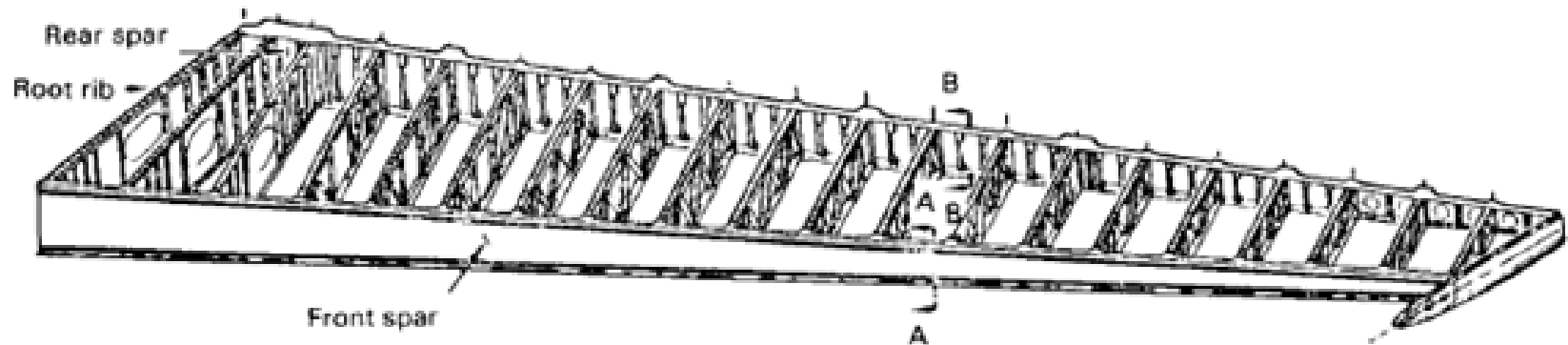
There are two basic configurations:

- Box constructions with spars, closer light rib spacing and covers without stringers reinforcements.

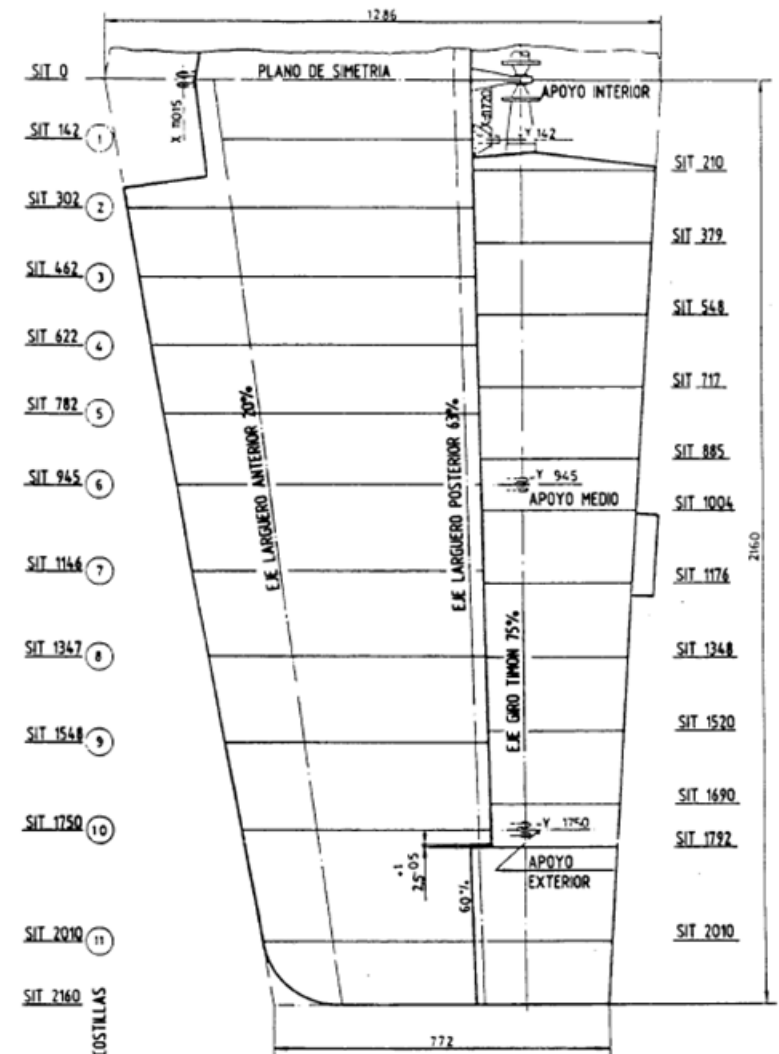
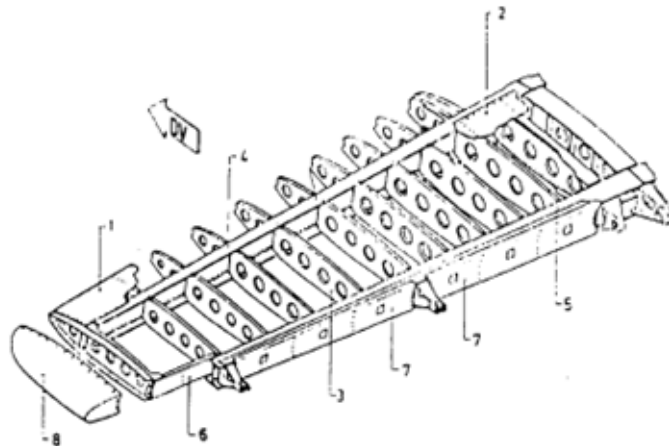


Transport Configurations

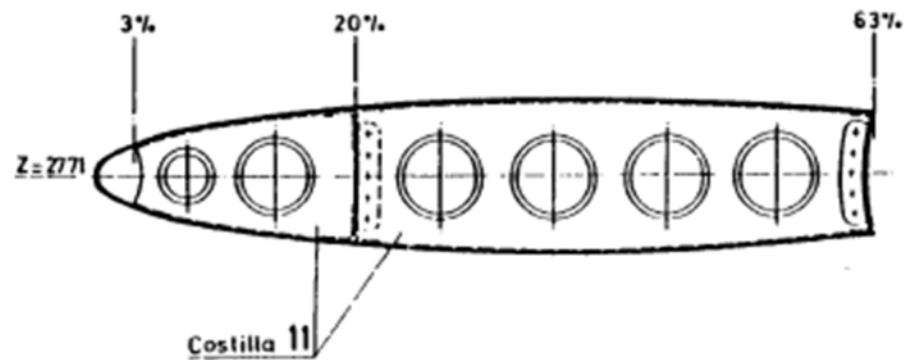
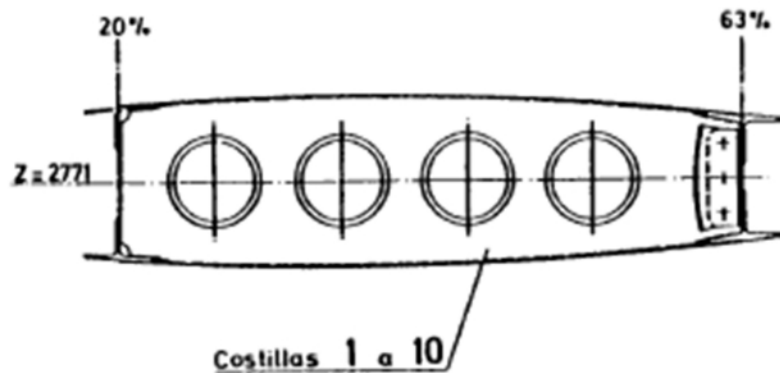
- Box construction with spars, stronger ribs and surface skins with stringer reinforcements is a lighter structure.



A technical drawing of a mechanical assembly, possibly a hinge or latch, shown in a perspective view. The assembly consists of two main parts, 1 and 2, which are joined together. Part 1 is a long, rectangular component with a textured surface. Part 2 is a smaller, cylindrical component that fits into a slot in part 1. The assembly is shown in a partially disassembled state, with part 2 being moved away from part 1. Numbered callouts (1, 2, 3, 4, 5, 6) point to various components and features of the assembly.

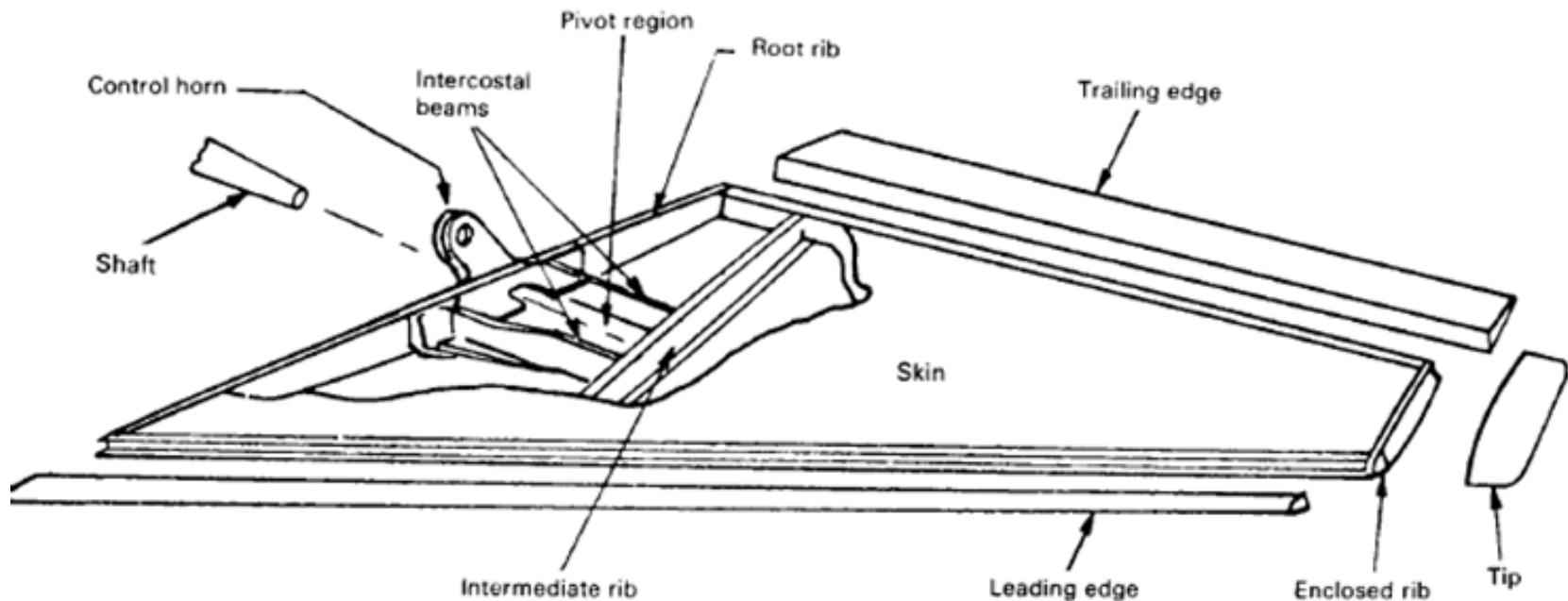


C101 Horizontal Estabilizer



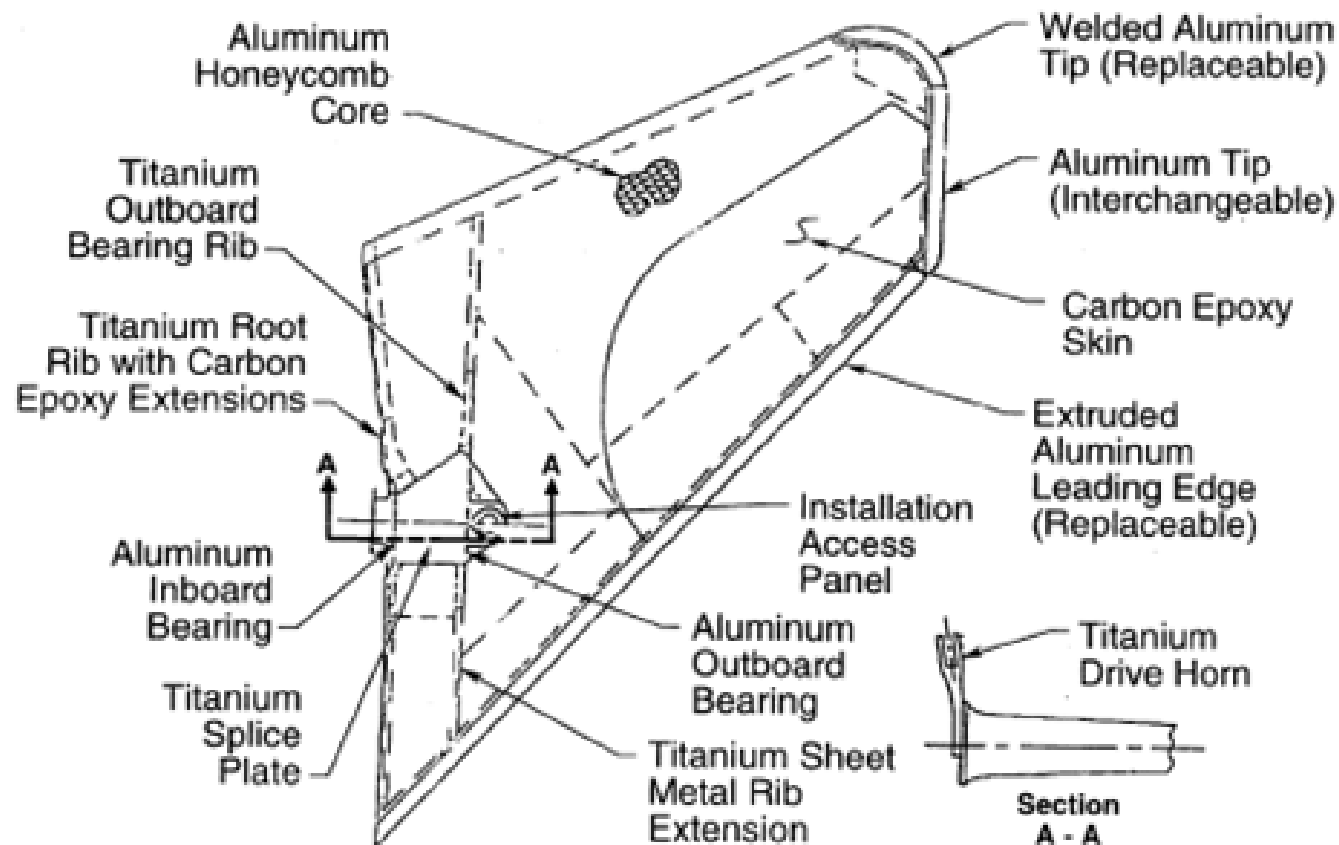
Fighter Configuration

Usually, on fighters, horizontal tail is complete movable and act as control surface.



F18

F-18 Stabilator Structural Arrangement



F18



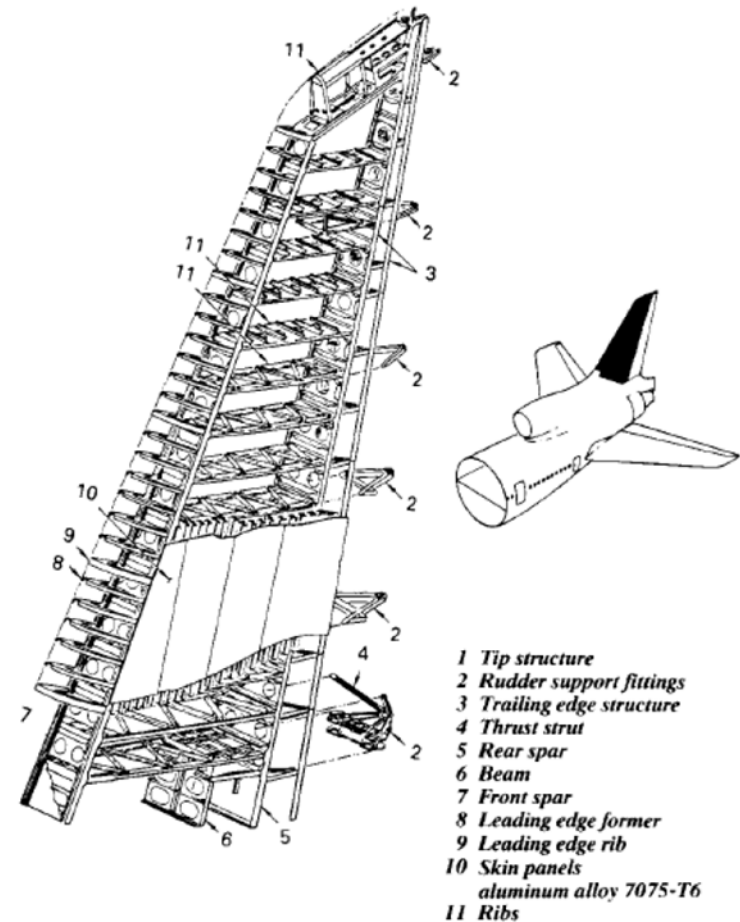
Vertical Tail

Typical Features

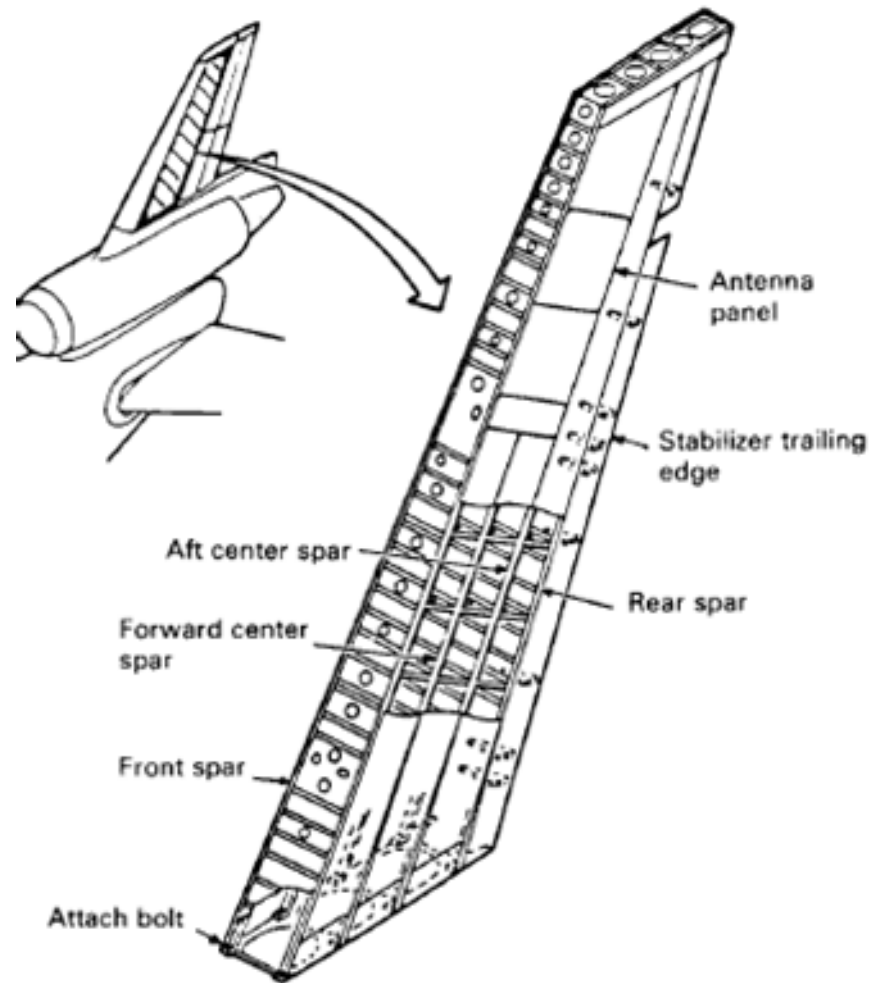
Structural Design of vertical stabilizers is essentially the same as for horizontal stabilizers.

Typical **Configurations** are:

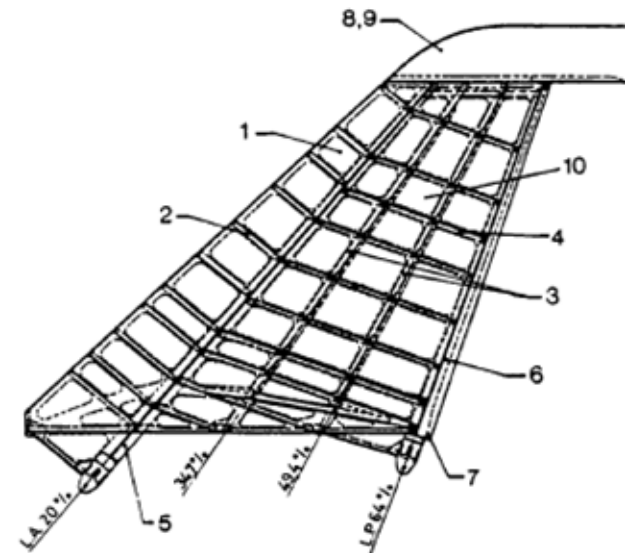
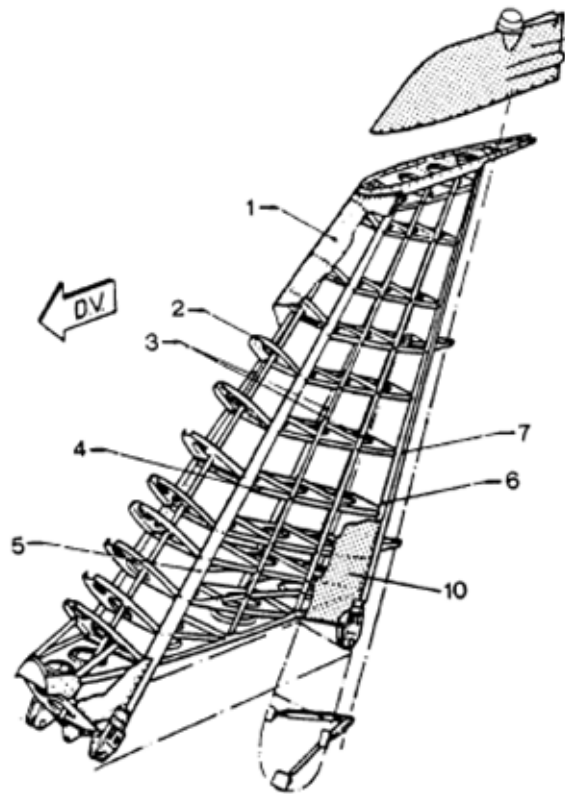
- Two spars with skin-stringer panels
- Multi-spars design
- Two spar, closer light rib spacing and skins without reinforcement stringers



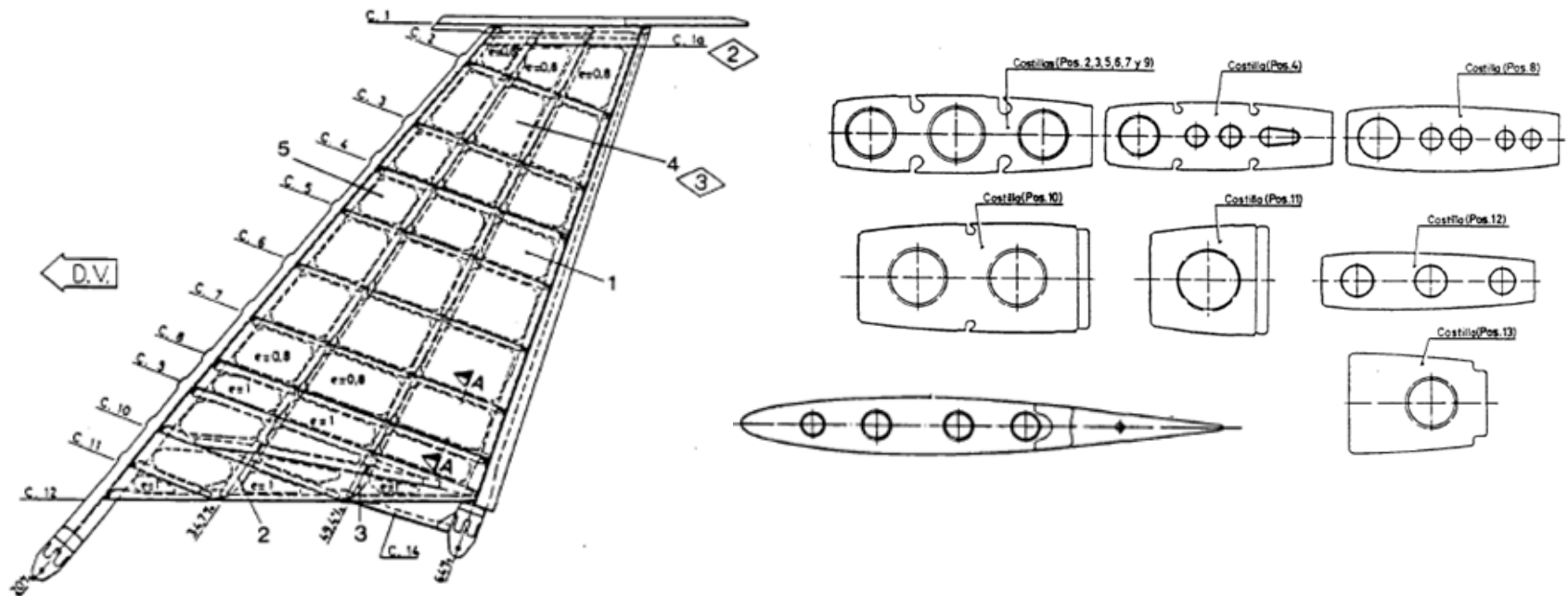
Typical Features



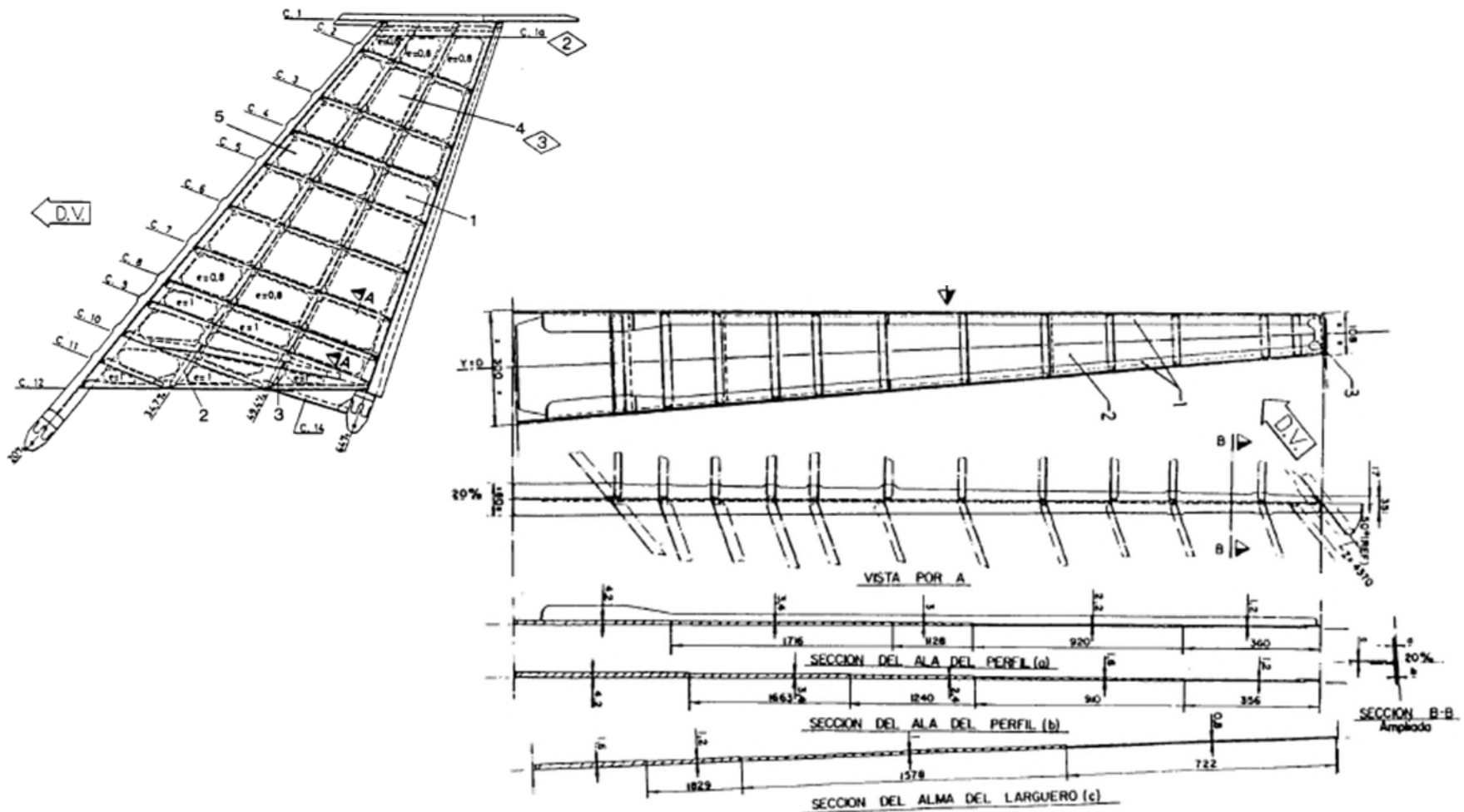
C101 Vertical Tail



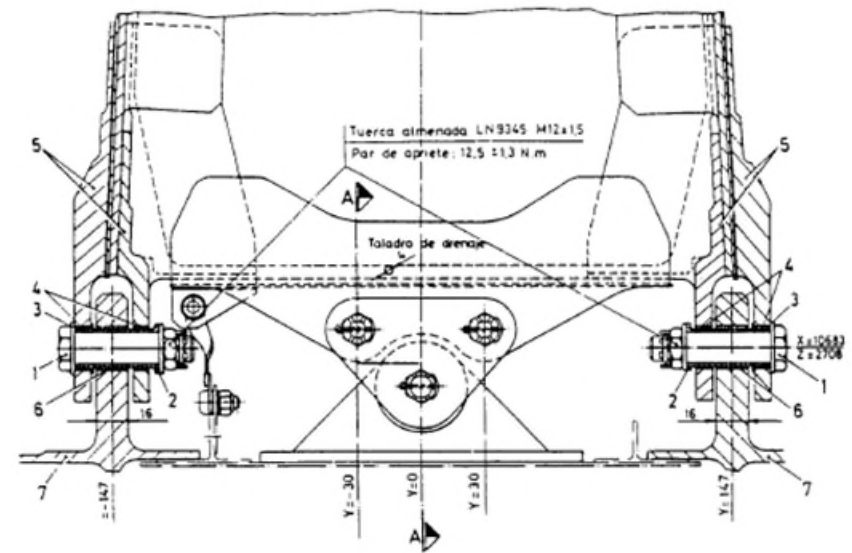
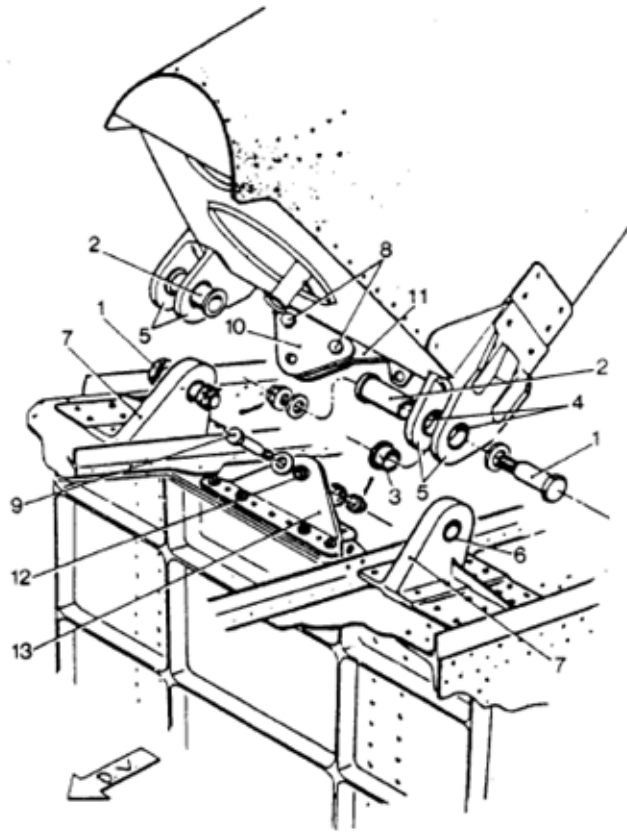
C101 Vertical Tail



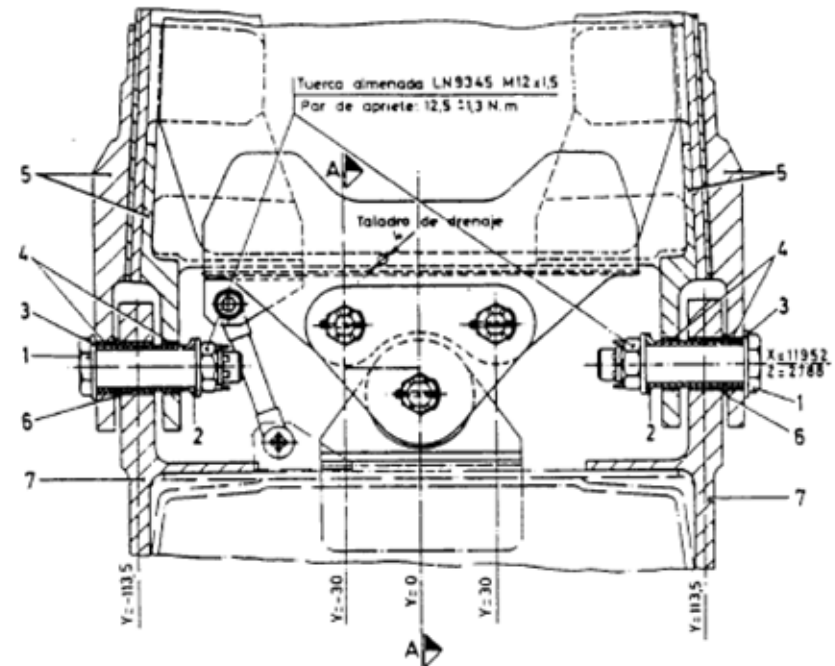
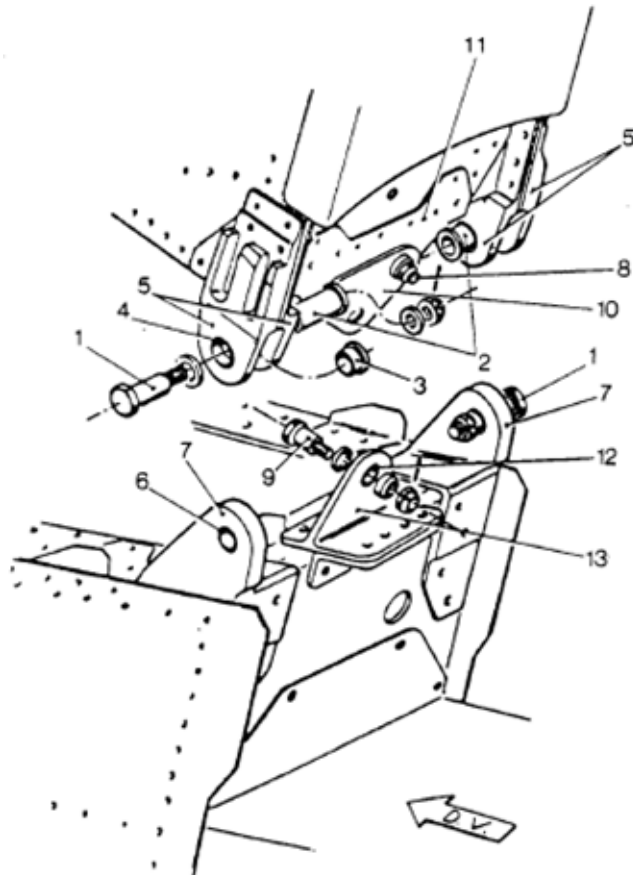
C101 Vertical Tail



C101 Vertical Tail Interface (@20%)



C101 Vertical Tail Interface (@64%)



1. Brunh, E. F., "Analysis and Design of Flight Vehicle Structures", 1973
2. Niu, M. C-Y, "Airframe Structural Design", 1988
3. Roskam, J., "Airplane Design", 1985
4. Torenbeek, E., "Synthesis of Subsonic Airplane Design", 1976
5. Airbus Defence and Space Internal Documentation